Semiconductor wafers — Solution

Y20 (2020) — English translation

A1 ?? pts

Obtain the values of I_{AB}, U_{CD}, I_{DA} and U_{BC} for 15 different connections to the plate.

A2 0.90 pts

Compute $R_{AB,CD}$, $R_{BC,DA}$ and y for each of the connections. Round all values to the nearest 4 significant figures.

A3 0.10 pts

The formula $\frac{y-1}{y+1} = \frac{1}{x}\text{arch}\left(\frac{e^x}{2}\right)$ can be converted to the form $e^x = e^{\alpha x} + e^{\beta x}$. Find α and β. Express your answer using y.

A4 3.00 pts

The equation $e^x = e^{\alpha x} + e^{\beta x}$ is transcendental and cannot be solved analytically. Solve it numerically, i.e. for each y, find e^x (to 4 significant figures).

$R_{AB,CD}$	$R_{BC,DA}$	y	e^x	x	$\frac{x}{y+1}$
$6{,}612 \cdot 10^5$	$3{,}805 \cdot 10^5$	1,738	2,037	0,7115	$2{,}599 \cdot 10^{-1}$
$9{,}335 \cdot 10^5$	$2{,}366 \cdot 10^5$	3,945	2,233	0,7989	$1{,}616 \cdot 10^{-1}$
$1{,}117 \cdot 10^6$	$1{,}793 \cdot 10^5$	6,231	2,425	0,8858	$1{,}225 \cdot 10^{-1}$
$1{,}469 \cdot 10^6$	$1{,}057 \cdot 10^5$	13,90	2,932	1,076	$7{,}221 \cdot 10^{-1}$
$1{,}793 \cdot 10^6$	$6{,}600 \cdot 10^4$	27,17	3,562	1,270	$4{,}508 \cdot 10^{-2}$
$1{,}917 \cdot 10^6$	$5{,}545 \cdot 10^4$	34,57	3,846	1,347	$3{,}787 \cdot 10^{-2}$
$2{,}099 \cdot 10^6$	$4{,}286 \cdot 10^4$	48,97	4,318	1,463	$2{,}928 \cdot 10^{-2}$
$2{,}175 \cdot 10^6$	$3{,}854 \cdot 10^4$	56,44	4,534	1,512	$2{,}632 \cdot 10^{-2}$
$2{,}301 \cdot 10^6$	$3{,}227 \cdot 10^4$	71,31	4,923	1,594	$2{,}204 \cdot 10^{-2}$
$2{,}580 \cdot 10^6$	$2{,}189 \cdot 10^4$	117,8	5,913	1,777	$1{,}496 \cdot 10^{-2}$
$2{,}975 \cdot 10^6$	$1{,}267 \cdot 10^4$	234,8	7,699	2,041	$8{,}656 \cdot 10^{-3}$
$3{,}278 \cdot 10^6$	$8{,}353 \cdot 10^3$	392,5	9,442	2,245	$5{,}705 \cdot 10^{-3}$
$3{,}493 \cdot 10^6$	$6{,}224 \cdot 10^3$	561,2	10,91	2,390	$4{,}251 \cdot 10^{-3}$
$3{,}689 \cdot 10^6$	$4{,}758 \cdot 10^3$	775,4	12,46	2,523	$3{,}250 \cdot 10^{-3}$
$3{,}808 \cdot 10^6$	$4{,}042 \cdot 10^3$	942,1	13,51	2,603	$2{,}760 \cdot 10^{-}$

A5 2.00 pts

Having plotted a graph in suitable coordinates, determine the numerical value ρ of the semiconductor. Round your answer to 3 significant figures. Take the thickness of the material equal to $d = 1$ mm.

$\rho = (2.3 \cdot 10^3 \pm 5)$

B1 ?? pts

Remove dependency $n(T)$.

$\ln n$	$\frac{1}{T}, \frac{1}{К}$	$\ln(nT^{\frac{-3}{4}})$	$\ln(nT^{\frac{-3}{2}})$
54,165	$3{,}33 \cdot 10^{-2}$	51,614	
54,745	$2{,}50 \cdot 10^{-2}$	51,978	
55,124	$2{,}00 \cdot 10^{-2}$	52,190	
55,405	$1{,}67 \cdot 10^{-2}$	52,335	
55,618	$1{,}43 \cdot 10^{-2}$	52,432	
55,796	$1{,}25 \cdot 10^{-2}$	52,510	
55,943	$1{,}11 \cdot 10^{-2}$	52,568	
56,068	10^{-2}	52,614	
56,176	$9{,}09 \cdot 10^{-3}$	52,651	
56,275	$8{,}33 \cdot 10^{-3}$	52,685	
56,357	$7{,}69 \cdot 10^{-3}$		
56,359	$4{,}98 \cdot 10^{-3}$		
56,359	$3{,}676 \cdot 10^{-3}$		
56,359	$2{,}915 \cdot 10^{-3}$		
56,359	$2{,}415 \cdot 10^{-3}$		
56,360	$1{,}590 \cdot 10^{-3}$		
58,666	$1{,}186 \cdot 10^{-3}$		48,561
58,724	$1{,}172 \cdot 10^{-3}$		48,601
58,782	$1{,}159 \cdot 10^{-3}$		48,642
58,839	$1{,}145 \cdot 10^{-3}$		48,681
58,896	$1{,}133 \cdot 10^{-3}$		48,721
58,951	$1{,}120 \cdot 10^{-3}$		48,760
59,005	$1{,}107 \cdot 10^{-3}$		48,797
59,056	$1{,}095 \cdot 10^{-3}$		48,834
59,111	$1{,}083 \cdot 10^{-3}$		48,869
59,162	$1{,}072 \cdot 10^{-3}$		48,905
59,213	$1{,}060 \cdot 10^{-3}$		48,939

B2) $T_i = (128 \pm 5)$К, $T_s = (780 \pm 40)$К
B3) $E_d = (1{,}18 \pm 0{,}02) \cdot 10^{-21}$ Дж
B4) $\Delta E_g = (8{,}23 \pm 0{,}11) \cdot 10^{-20}$Дж

B2 1.60 pts

Plot the dependence $\ln n\ (\frac{1}{T})$. Approximate three sections of this graph with straight lines and determine the values of T_i and T_s. Estimate the error of the obtained values.

B3 1.20 pts

Find E_d.

B4 1.20 pts

Find ΔE_g.